

Mark schemes

Q1.

(a) Work done = $\int_0^e \frac{\lambda x}{l} dx$

M1

$$= \left[\frac{\lambda x^2}{2l} \right]_0^e$$

Needs limit of 0

A1

$$= \frac{\lambda e^2}{2l}$$

A1

Or

area under a straight line
= average force $\times$ distance

$$= \frac{\lambda e^2}{2l}$$

3

(b) (i) Using $T = \frac{\lambda x}{l}$
 $5g = \frac{150 \times x}{0.6}$

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M1

Extension is 0.196 m

A1

2

(ii) $EPE = \frac{\lambda x^2}{2l}$
 $= \frac{150 \times (1.4)^2}{2 \times 0.6}$

M1

$$= 245 \text{ J}$$

A1

2

- (iii) When at point O ,
 $EPE = 0$
 $PE \text{ [relative to } P] = 5 \times g \times 2$

M1

$$KE \text{ [at } O] = EPE \text{ [at } P] - \text{gain in PE [at } O]$$

M1

$$\frac{1}{2}mv^2 = 245 - 10g$$

$$\frac{1}{2} \cdot 5 \cdot v^2 = 147$$

A1

$$v^2 = 58.8 \quad \text{Speed is } 7.67 \text{ m s}^{-1}$$

A1

4

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Q2.

(a) Work done = $\int_0^e \frac{\lambda x}{l} dx$

M1

$$= \left[\frac{\lambda x^2}{2l} \right]_0^e$$

Needs limit of 0

A1

$$= \frac{\lambda e^2}{2l}$$

AG

A1

Or

$$\text{area under a straight line} = \text{average force} \times \text{distance} = \frac{\lambda e^2}{2l}$$

3

(b) (i) Using $T = \frac{\lambda x}{l}$

$$5g = \frac{150 \times x}{0.6}$$

M1

Extension is 0.196 m

A1

2

$$(ii) \quad EPE = \frac{\lambda x^2}{2l}$$

$$= \frac{150 \times (0.3)^2}{2 \times 0.6}$$

M1

$$= 11.25 \text{ J}$$

A1

2

$$(iii) \quad \text{When } x \text{ above } P,$$

$$EPE = \frac{150 \times (0.3 - x)^2}{2 \times 0.6} =$$

$$\text{for } \frac{150 \times (\dots - x)^2}{2 \times 0.6}$$

M1A1

$$PE [\text{relative to } P] = (-) 5 \times g \times x$$

for $5 \times g \times \text{distance}$

M1

$$KE + EPE [\text{at new point}]$$

$$= EPE [\text{at } P] - \text{gain in PE}$$

4 terms, all signs correct, 2 terms correct

M1

$$\frac{1}{2}mv^2 + \frac{150 \times (0.3 - x)^2}{2 \times 0.6} = \frac{150 \times (0.3)^2}{2 \times 0.6} - 5gx$$

A1

$$\frac{1}{2}mv^2 + \frac{150 \times (x^2 - 0.6x)}{2 \times 0.6} = -5gx$$

Equation involving terms in v^2 , x^2 and x only

m1

$$\frac{1}{2} \cdot 5 \cdot v^2 + 125x^2 - 75x = -49x$$

$$v^2 = 10.4x - 50x^2$$

A1

7

- (iv) Particle is at rest when $v = 0$
 $10.4x - 50x^2 = 0$

M1

$$x = 0 \text{ [not required]}$$

$$\text{Or } x = \frac{10.4}{50} = 0.208 \text{ m above } P.$$

A1

2

[16]

EXAM PAPERS PRACTICE

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Examiner reports

Q1.

Part (a) tested work done = $\int F dx$. Few candidates found $\int_0^e \frac{\lambda x}{l} dx$ correctly. Instead of integrating, a few candidates used the value of the integral to be the area under the line $y = \frac{\lambda x}{l}$. Unfortunately, many candidates used techniques which were not credited: for

example, elastic potential energy is $\frac{\lambda x^2}{2l}$ and $x = e$; or work done = maximum force $\times$ half the distance moved, without a clear justification.

Part (b)(i) and (ii) were completed well. In part (b)(iii), most candidates attempted to conserve kinetic energy, potential energy and elastic potential energy, but often the signs

of the three terms in the equation $\frac{1}{2}mv^2 - 245 + 10g = 0$ were incorrect.

Q2.

Part (a) tested work done = $\int F dx$. Few candidates found $\int_0^e \frac{\lambda x}{l} dx$ correctly. Instead of integrating, a few candidates used the value of the integral to be the area under the line $y = \frac{\lambda x}{l}$. Unfortunately, many candidates used techniques which were not credited: for

example, elastic potential energy is $\frac{\lambda x^2}{2l}$ and $x = e$; or work done = maximum force $\times$ half the distance moved, without a clear justification.

Part (b)(i) was completed well, as was part (b)(ii), although a number of candidates did

write down $\frac{150 \times 0.9^2}{2 \times 1.2} = 11.25$, which evidently it is not. In part (b)(iii), many tried to use the conservation of kinetic energy, potential energy and elastic potential energy. The printed result enabled many of those who obtained an incorrect answer to correct their

working. Commonly, this was because they had used $\frac{\lambda(0.9 - x)^2}{2l}$ instead of $\frac{\lambda(0.3 - x)^2}{2l}$ for the elastic potential energy. Part (c) was completed well, but some candidates did not explicitly exclude $x = 0$.